Assignment 6 (Recursion)

Solve all 10. For "write a function" questions, make sure your base case and recursive case are both clear. For "output" questions, work out the order by hand before running anything. Show your working where asked.

Easy

Q1. What is the exact output of this program?

```python
def greet(n):
    if n == 0:
        return
    print("Namaste")
    greet(n - 1)

greet(3)
```

Q2. Write a recursive function `count_up(n)` that prints the numbers from `1` to `n`, each on its own line. Example: `count_up(4)` should print `1`, `2`, `3`, `4` (in that order).

Q3. In the function below, write down which line is the **base case** and which line is the **recursive case**, and explain in one sentence what the base case is for.

```python
def fact(n):
    if n == 0:
        return 1
    return n * fact(n - 1)
```

Q4. What value does this program print?

```python
def total(n):
    if n == 0:
        return 0
    return n + total(n - 1)

print(total(5))
```

Medium

Q5. What is the exact output of this program? (Watch *where* the `print` sits.)

```python
def show(n):
    if n == 0:
        return
    show(n - 1)
    print(n)

show(4)
```

Q6. Write a recursive function `digit_sum(n)` that returns the sum of the digits of a non-negative integer. Example: `digit_sum(2025)` should return `9` (because `2 + 0 + 2 + 5 = 9`).

Q7. The function below is meant to return the sum of all numbers in a list, but it always returns `0`. Find the bug, explain why it happens, and write the corrected function.

```python
def list_sum(a):
    if not a:
        return 0
    return list_sum(a[1:])
```

Q8. Using the GCD function below, what does `gcd(36, 24)` return? Show the full sequence of calls (for example, `gcd(36, 24) → gcd(24, 12) → ...`).

```python
def gcd(a, b):
    if b == 0:
        return a
    return gcd(b, a % b)
```

Hard

Q9. Write a recursive function `is_palindrome(s)` that returns `True` if a string reads the same forwards and backwards, and `False` otherwise. Examples: `is_palindrome("NITIN")` → `True`; `is_palindrome("INDIA")` → `False`. *(State your base case clearly.)*

Q10. Write a recursive function `power(base, exp)` that computes `base` raised to the power `exp` (assume `exp` is `0` or a positive integer). Example: `power(3, 4)` should return `81`. Then answer: **how many times is `power` called in total** (including the first call) when you evaluate `power(3, 4)`? Explain your count.